Robust Estimation of Graph Statistics

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1 Introduction

1.1 Background

1.2 Structure of this paper

1.3 Notations

Definition 1.1. *A graph $G \sim \mathcal{G}(n, d, \epsilon)$ is a balanced, two-community stochastic block model graph with $V = [n]$ and uniformly random label vector $\boldsymbol{x} \in \{\pm 1\}^n$ satisfying $\mathbf{1}^\top \boldsymbol{x} = 0$. For each pair of vertices $i, j \in [n]$, we have*

$$\mathbb{P}((i,j) \in E) = \begin{cases} \frac{d}{n}(1 + \frac{\epsilon}{2}) & \text{if } x_i = x_j, \\ \frac{d}{n}(1 - \frac{\epsilon}{2}) & \text{if } x_i \neq x_j. \end{cases}$$

Definition 1.2. *The set of positive semidefinite matrices with diagonal entries equal to 1 is denoted by*

$$\mathcal{PSD}_1(n) = \{\boldsymbol{X} \in \mathbb{R}^{n \times n} : \boldsymbol{X} \succeq 0, \boldsymbol{X}_{ii} = 1 \text{ for all } i \in [n]\}.$$

Definition 1.3. *Given a matrix $\boldsymbol{M} \in \mathbb{R}^{n \times n}$, we define $\mathcal{SDP}(\boldsymbol{M}) := \max_{\boldsymbol{X} \in \mathcal{PSD}_1(n)} \langle \boldsymbol{M}, \boldsymbol{X} \rangle$.*

Definition 1.4. *Given a graph G, let $\boldsymbol{A}_G$ be its adjacency matrix, and define $\boldsymbol{A}_G^{\text{cent}} = \boldsymbol{A}_G - \mathbb{E}[\boldsymbol{A}_G]$. For $G \sim \mathcal{G}(n, d, \epsilon)$, we have $\mathbb{E}[\boldsymbol{A}_G] = \frac{d}{n}\mathbf{1}\mathbf{1}^\top$.*

2 (Looking at MS16)

introduce MS16

2.1 Algorithm

Algorithm 1 (SDP Algorithm from [MS16])**.**

1. Sample edges of G independently with probability $(1 + n^{-1/2})^{-1}$ to obtain the sampled edge set $E_{\text{sample}} \subset E$, and then compute the evaluation edge set $E_{\text{eval}} = E \setminus E_{\text{sample}}$.
2. Construct the subgraphs $G_{\text{sample}} = (V, E_{\text{sample}})$ and $G_{\text{eval}} = (V, E_{\text{eval}})$.
3. Compute an optimiser $\boldsymbol{X}^*$ of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$, i.e., $\langle \boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \rangle = \mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$.
4. Compute the eigenvectors $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(n)}$ of $\boldsymbol{X}^*$.
5. For each $i, j \in [n]$, define $\hat{\boldsymbol{x}}^{(i,j)} \in \{-1, 0, 1\}^n$ as follows:

$$\hat{\boldsymbol{x}}_k^{(i,j)} = \begin{cases} \operatorname{sign}(\boldsymbol{v}_k^{(i)}) & \text{if } |\boldsymbol{v}_k^{(i)}| \geq |\boldsymbol{v}_j^{(i)}|, \\ 0 & \text{otherwise.} \end{cases}$$

6. Find $(i^*, j^*) = \arg\max_{i,j \in [n]} (\hat{\boldsymbol{x}}^{(i,j)})^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(i,j)} = \sum_{a,b \in E_{\text{eval}}} \hat{\boldsymbol{x}}_a^{(i,j)} \cdot \hat{\boldsymbol{x}}_b^{(i,j)}$.

7. Output $\hat{\boldsymbol{x}}^{\mathrm{SDP}}(G) = \hat{\boldsymbol{x}}^{(i^*,j^*)}$.

We have the following weak recovery rate guarantee for Algorithm 1.

Theorem 2.1 (Weak Recovery for Algorithm 1)**.** *Let $G \sim \mathcal{G}(n, d, \epsilon)$ and assume, for some $\delta > 0$, that $\lambda = \epsilon\sqrt{d}/2 \geq 1 + \delta$. Then, there exists $\eta = \eta(\delta) > 0$ and $d_* = d_*(\delta) > 0$ such that, for all $d \geq d_*$, the output of Algorithm 1, $\hat{\boldsymbol{x}}^{\mathrm{SDP}}(G)$, satisfies*

$$\mathbb{P}\left(\frac{1}{n}|\langle \hat{\boldsymbol{x}}^{\mathrm{SDP}}(G), x\rangle| \geq \eta\right) \geq 1 - Ce^{-\sqrt{n}/C}.$$

discuss robustness of the algorithm

2.2 Proof of Weak Recovery (Theorem 2.1)

Let $\boldsymbol{x}$ be the label vector of $G \sim \mathcal{G}(n, d, \epsilon)$. Note that $G_{\mathrm{sample}} \sim \mathcal{G}(n, d_{\mathrm{sample}}, \epsilon)$, and $G_{\mathrm{eval}} \sim \mathcal{G}(n, d_{\mathrm{eval}}, \epsilon)$, where $d_{\mathrm{sample}} = d/(1 + n^{-1/2})$ and $d_{\mathrm{eval}} = n^{-1/2} d_{\mathrm{sample}}$. For simplicity, from now on we write $d = d_{\mathrm{sample}}$.

introduce the proof strategy

2.2.1 Step 1: $\boldsymbol{X}^*$ contains nontrivial information about $\boldsymbol{x}$

We first introduce a group of Gaussian random matrices (GOEs) that have well-behaved SDP values.

investigate independence

Definition 2.1 (Gaussian orthogonal ensemble (GOE) and the deformed GOE $\boldsymbol{B}(\lambda)$)**.** *A random matrix $\boldsymbol{W} \in \mathbb{R}^{n\times n}$ is a* GOE matrix *if it is a symmetric matrix with $\{\boldsymbol{W}_{ij}\}_{i\leq j}$ independent and $\boldsymbol{W}_{ij} \sim \mathcal{N}(0, 1/n)$ for $i < j$ and $\boldsymbol{W}_{ii} \sim \mathcal{N}(0, 2/n)$. The* deformed GOE matrix *$\boldsymbol{B}(\lambda)$ is then defined as*

$$\boldsymbol{B}(\lambda) = \frac{\lambda}{n}\mathbf{1}\mathbf{1}^\top + \boldsymbol{W}.$$

We then have the following phase transition result for the SDP value of $\boldsymbol{B}(\lambda)$.

Lemma 2.1 (SDP value of $\boldsymbol{B}(\lambda)$, Theorem 5 of [MS16])**.** *Let $\boldsymbol{B}(\lambda)$ be a deformed GOE matrix defined in Definition 2.1. Then, with probability converging to 1 as $n \to \infty$,*

(a) if $\lambda \leq 1$: for any $\epsilon > 0$, we have $\mathcal{SDP}(\boldsymbol{B}(\lambda))/n \in [2 - \epsilon, 2 + \epsilon]$.

(b) if $\lambda > 1$: there exists $\Delta(\lambda) > 0$ such that $\mathcal{SDP}(\boldsymbol{B}(\lambda))/n \geq 2 + \Delta(\lambda)$.

If we set the diagonal entries of $\boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{sample}}}$ to be $\lambda\sqrt{d}/n$, then

$$\frac{1}{\sqrt{d}}\boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{sample}}} = \frac{\lambda}{n}\boldsymbol{x}\boldsymbol{x}^\top + \boldsymbol{E},$$

where $\boldsymbol{E}$ is a symmetric matrix with zero diagonal and independent off-diagonal entries satisfying

$$\boldsymbol{E}_{ij} = \begin{cases} \frac{1}{\sqrt{d}} - \frac{\sqrt{d}}{n} - \frac{\lambda}{n}\boldsymbol{x}_i\boldsymbol{x}_j & \text{with probability } \mathbb{P}((i,j) \in E_{\mathrm{sample}}), \\ -\frac{\sqrt{d}}{n} - \frac{\lambda}{n}\boldsymbol{x}_i\boldsymbol{x}_j & \text{with probability } 1 - \mathbb{P}((i,j) \in E_{\mathrm{sample}}). \end{cases}$$

Let $\boldsymbol{X}^*$ be an optimizer of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$. We will now try to bound $\langle \boldsymbol{E}, \boldsymbol{X}^* \rangle$ and $\langle \frac{1}{\sqrt{d}} \boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \rangle$ away from each other using the phase transition result of Lemma 2.1, which will then give us a lower bound on $n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle$, which is the correlation between $\boldsymbol{X}^*$ and the ground truth labels matrix $\boldsymbol{x}\boldsymbol{x}^\top$.

The following lemma bounds $\mathcal{SDP}(\boldsymbol{E})$ to the SDP value of a GOE matrix.

Lemma 2.2 (GOE approximation of $\mathcal{SDP}(\boldsymbol{E})$)**.** *Let $\boldsymbol{W}$ be a GOE matrix. Then, there exists $C = C(\lambda) > 0$ such that, for all $n \geq n_0(d, \epsilon)$ with probability at least $1 - Ce^{-n/C}$, we have*

$$\left| \frac{1}{n}\mathcal{SDP}(\boldsymbol{E}) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{W}) \right| \leq \frac{C\log(d)}{d^{1/10}}.$$

Moreover, $C(\lambda)$ is bounded for λ in any compact set.

We also have a similar result bounding $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}/\sqrt{d})$ by the SDP value of a deformed GOE matrix.

Lemma 2.3 (Deformed GOE approximation of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}/\sqrt{d})$)**.** *Let $\boldsymbol{B}(\lambda)$ be a deformed GOE matrix. Then, there exists $C = C(\lambda) > 0$ such that, for all $n \geq n_0(d, \epsilon)$*

$$\left| \frac{1}{n}\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}\right) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{B}(\lambda)) \right| \leq \frac{C\log(d)}{d^{1/10}}.$$

Moreover, $C(\lambda)$ is bounded for λ in any compact set.

Now we have the tools to obtain a lower bound on $n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle$.

$$\begin{aligned}
\left\langle \frac{\lambda}{n^2}\boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \right\rangle &= \frac{1}{n}\left(\left\langle \frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \right\rangle - \langle \boldsymbol{E}, \boldsymbol{X}^* \rangle \right) \\
&= \frac{1}{n}\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}\right) - \frac{1}{n}\langle \boldsymbol{E}, \boldsymbol{X}^* \rangle \\
&\geq \frac{1}{n}\mathcal{SDP}(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{E}) && \text{by optimality of SDP} \\
&\geq \frac{1}{n}\mathcal{SDP}(\boldsymbol{B}(\lambda)) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{W}) - \frac{C\log(d)}{d^{1/10}} && \text{by Lemmas 2.2 and 2.3} \\
&\geq \Delta_1(\lambda) - \frac{C\log(d)}{d^{1/10}}, && \text{by Lemma 2.1}
\end{aligned}$$

with probability at least $1 - Cn^{-4}$ (see the remark below).

Remark 2.1. In the original paper [MS16], the authors claimed that the above inequality holds with probability at least $1 - Ce^{-n/C}$, but tracking the proof of Lemma 2.1 (Appendices F and G of [MS16]) shows that they only proved the result with probability at least $1 - Cn^{-4}$ (Lemma G.8 of [MS16]). This is still sufficient for our purposes, as the probability still tends to 1 as $n \to \infty$.

Now fix λ, and take d sufficiently large such that $\Delta_1(\lambda) - \frac{C\log(d)}{d^{1/10}} \geq \lambda\Delta_2(\lambda) > 0$. Then, we have, with probability at least $1 - Cn^{-4}$,

$$n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle \geq \Delta_2(\lambda).$$

This is an important milestone, as it shows that $\boldsymbol{X}^*$ has a nontrivial correlation with the ground truth $\boldsymbol{x}\boldsymbol{x}^\top$.

2.2.2 Step 2: There exists a good label estimator among $\{\hat{\boldsymbol{x}}^{(i,j)}\}_{i,j\in[n]}$

We first define the following rounding function used in step 5 of Algorithm 1:

Definition 2.2 (Rounding function)**.** *For $t \geq 0$, define the* rounding function $r_t : \mathbb{R} \to \{-1, 0, +1\}$ *as follows:*

$$r_t(x) = \begin{cases} 1 & \text{if } x \geq t, \\ 0 & \text{if } -t < x < t, \\ -1 & \text{if } x \leq -t. \end{cases}$$

Hence, we can write $\hat{\boldsymbol{x}}^{(\hat{\imath},j)} = r_{\boldsymbol{v}_j^{(\hat{\imath})}}(\boldsymbol{v}^{(\hat{\imath})})$.

Since $\boldsymbol{X} \in \mathcal{PSD}_1(n)$, writing its eigen-decomposition as $\boldsymbol{X} = \sum_{i=1}^n \xi_i \boldsymbol{v}^{(i)}(\boldsymbol{v}^{(i)})^\top$, where ξ_i is the i-th largest eigenvalue, and $\boldsymbol{v}^{(i)}$ is the corresponding unit length eigenvector, we have $\xi_i \geq 0$ for all i by positive semidefiniteness, and $\sum_{i=1}^n \xi_i = \mathrm{Tr}(\boldsymbol{X}) = n$. Thus,

$$\Delta_2(\lambda) \leq \frac{1}{n^2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^*\rangle = \frac{1}{n^2}\sum_{i=1}^n \xi_i(\boldsymbol{x}^\top \boldsymbol{v}^{(i)})^2 = \sum_{i=1}^n \frac{\xi_i}{n} \cdot \left(\frac{\boldsymbol{x}^\top \boldsymbol{v}^{(i)}}{\sqrt{n}}\right)^2.$$

So there exists $\hat{\imath} \in [n]$ such that $|\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})}| \geq \sqrt{n\Delta_2(\lambda)}$.

Next, we will use following lemma to justify that the rounding function preserves this correlation, which shows that it is possible to extract a good label estimator.

Lemma 2.4 (Rounding Lemma)**.** *Let X, Y be random variables such that: $\mathbb{P}(X = +1) = \mathbb{P}(X = -1) = 1/2$, $\mathbb{E}(XY) \geq \epsilon > 0$, and $\mathbb{E}(Y^2) = 1$. Then, there exists t^* depending on the joint distribution of X and Y, such that*

$$\mathbb{E}(X r_{t^*}(Y)) \geq \frac{\epsilon^2}{2}.$$

Proof. Note that for all t, we have $X r_t(Y) = r_t(XY)$, and $Y^2 = (XY)^2$. Let $Z = XY$, then it suffices to show that given $\mathbb{E}(Z) \geq \epsilon > 0$ and $\mathbb{E}(Z^2) = 1$, we have $\mathbb{E}(r_t(Z)) \geq \epsilon^2/2$. Recall that for nonnegative random variable S, by Fubini, we have

$$\mathbb{E}(S) = \mathbb{E}\left(\int_0^\infty \mathbb{1}_{s\leq S} ds\right) = \int_0^\infty \mathbb{P}(S \geq s) ds, \quad \mathbb{E}(S^2) = \mathbb{E}\left(\int_0^\infty 2s\mathbb{1}_{s\leq S} ds\right) = \int_0^\infty 2s\mathbb{P}(S \geq s) ds.$$

Therefore, writing $Z^{\pm} = \max(\pm Z, 0)$, we have

$$\begin{aligned}
\epsilon \le \mathbb{E}(Z) = \mathbb{E}(Z_+) - \mathbb{E}(Z_-) &= \int_0^\infty \mathbb{P}(Z_+ \ge z)dz - \int_0^\infty \mathbb{P}(Z_- \ge z)dz \\
&\le \int_0^K (\mathbb{P}(Z \ge z) - \mathbb{P}(Z \le -z))dz + \int_K^\infty \frac{2z}{2K}\mathbb{P}(Z \ge z)dz - \int_K^\infty \frac{2z}{2K}\mathbb{P}(-Z \ge z)dz \\
&= \int_0^K \mathbb{E}r_t(Z)dt + \frac{1}{2K}\mathbb{E}(Z_+^2 - Z_-^2) = \int_0^K \mathbb{E}r_t(Z)dt + \frac{1}{2K}.
\end{aligned}$$

Now take $K = 1/\epsilon$, then

$$\frac{1}{K}\int_0^K \mathbb{E}r_t(Z)dt \ge \frac{1}{K}(\epsilon - \frac{1}{2K}) = \frac{\epsilon^2}{2}$$

So the average of $\mathbb{E}r_t(Z)$ over $t \in [0, 1/\epsilon]$ is greater than $\epsilon^2/2$, so there exists $t^* \in [0, 1/\epsilon]$ such that $\mathbb{E}r_{t^*}(Z) \ge \epsilon^2/2$. □

Without loss of generality, assume that $\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})} > 0$. Consider (X, Y) drawn uniformly from $\{(\boldsymbol{x}_k, \sqrt{n}\boldsymbol{v}_k^{(\hat{\imath})})\}_{k\in[n]}$, then we have $\mathbb{E}(XY) = n^{-1/2}\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})} \ge \sqrt{\Delta_2(\lambda)}$ and $\mathbb{E}(Y^2) = n^{-1}||\sqrt{n}\boldsymbol{v}^{(\hat{\imath})}||^2 = ||\boldsymbol{v}^{(\hat{\imath})}||^2 = 1$. Thus, by Lemma 2.4, there exists $t^* \in [0, 1/\sqrt{\Delta_2(\lambda)}]$ such that

$$\frac{\Delta_2(\lambda)}{2} \le \mathbb{E}(Xr_{t^*}(Y)) = \frac{1}{n}\sum_{k=1}^n \boldsymbol{x}_k r_{t^*}(\sqrt{n}\boldsymbol{v}_k^{(\hat{\imath})}) = \frac{1}{n}\langle \boldsymbol{x}, r_{\tilde{t}}(\boldsymbol{v}^{(\hat{\imath})})\rangle,$$

where $\tilde{t} = t^*/\sqrt{n} \in [0, 1/\sqrt{n\Delta_2(\lambda)}]$.

Remark 2.2. Here, we slightly improved the rounding lemma bound from $\epsilon^2/4$ in [MS16] to $\epsilon^2/2$, although the main proof idea remains the same. Also, the original application of the rounding lemma missed the scaling factor $\sqrt{n}$ in the definition of Y, but the conclusion remains valid.

Finally, note that for any value of $\tilde{t}$, we can find $\hat{\jmath} \in [n]$ such that $r_{\tilde{t}}(\boldsymbol{v}^{(\hat{\imath})}) = r_{\boldsymbol{v}_{\hat{\jmath}}^{(\hat{\imath})}}(\boldsymbol{v}^{(\hat{\imath})}) = \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}$. Therefore, there exists some $\hat{\imath}, \hat{\jmath} \in [n]$ such that

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}\rangle \ge \frac{\Delta_2(\lambda)}{2}.$$

Now, it remains to show that the selection step (Step 6) of Algorithm 1 will find such $\hat{\imath}, \hat{\jmath}$ with high probability, which will then complete the proof of Theorem 2.1.

2.2.3 Step 3: The evaluation edge set gives a good estimator

Recall the well-known Azuma-Hoeffding inequality. add reference

Lemma 2.5 (Azuma-Hoeffding inequality)**.** *Let $X_0, \ldots, X_m$ be a martingale, with bounded increments, i.e. $|X_i - X_{i-1}| \le c_i$ for all $i \in [m]$. Then, for all $t > 0$, we have*

$$\mathbb{P}(|X_m - \mathbb{E}(X_m)| \ge t) \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^m c_i^2}\right).$$

We apply this inequality to the objective function in Step 6 of Algorithm 1, and obtain the following concentration result.

Lemma 2.6. *Let $G \sim \mathcal{G}(n, d, \epsilon)$. For any fixed vector $\hat{\boldsymbol{x}} \in \{-1, 0, +1\}^n$, define $f(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G^{\mathrm{cent}} \hat{\boldsymbol{x}}$. Then, for any $t > 0$,*

$$\mathbb{P}\Big(\big|f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}}))\big| \geq t \;\Big|\; |E_{\mathrm{eval}}| \leq m\Big) \leq 2\exp\left(-\frac{t^2}{8m}\right).$$

Proof. Note that $f(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G^{\mathrm{cent}} \hat{\boldsymbol{x}} = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G \hat{\boldsymbol{x}} - (d/n)(\hat{\boldsymbol{x}}^\top \mathbf{1})^2$, so writing $\tilde{f}(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G \hat{\boldsymbol{x}}$, we have $f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}})) = \tilde{f}(G; \hat{\boldsymbol{x}}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}))$. Note that $\tilde{f}(G; \hat{\boldsymbol{x}}) = \sum_{(a,b) \in E} 2\hat{\boldsymbol{x}}_a \hat{\boldsymbol{x}}_b$.

For some ordering of all possible edges, we can define a filtration $\mathcal{F}_0, \mathcal{F}_1, \ldots, \mathcal{F}_m$ with $\mathcal{F}_k = \sigma(E_1, \ldots, E_k)$, where $E_k \in E(G) \cup \varnothing$ is the k-th edge of G in the ordering. Then, define the martingale $X_k = \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | \mathcal{F}_k)$ for $k \in [m]$. Since $|E(G)| \leq m$, we have $X_m = \tilde{f}(G; \hat{\boldsymbol{x}})$. Note that for all $k \in [m]$, we have

$$\begin{aligned}
|X_k - X_{k-1}| &= \Big|\mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1})\Big| \\
&\leq \sup_{e_{1:k-1}, e_k, e_k'} \Big|\mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1} = e_{1:k-1}, E_k = e_k) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1} = e_{1:k-1}, E_k = e_k')\Big| \\
&\leq \sup_{E(G^+) = E(G) \cup \{(a,b)\}} \Big|\tilde{f}(G^+; \hat{\boldsymbol{x}}) - \tilde{f}(G; \hat{\boldsymbol{x}})\Big| \\
&\leq |2\hat{\boldsymbol{x}}_a \hat{\boldsymbol{x}}_b| \leq 2.
\end{aligned}$$

Hence, we can apply the Azuma-Hoeffding inequality Lemma 2.5 to X with $c_i = 2$ for all $i \in [m]$ to obtain

$$\begin{aligned}
\mathbb{P}\Big(\big|f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}}))\big| \geq t \;\Big|\; |E(G)| \leq m\Big) &= \mathbb{P}\Big(\big|\tilde{f}(G; \hat{\boldsymbol{x}}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}))\big| \geq t \;\Big|\; |E(G)| \leq m\Big) \\
&= \mathbb{P}(|X_m - \mathbb{E}(X_m)| \geq t) \\
&\leq 2\exp\left(-\frac{t^2}{8m}\right).
\end{aligned}$$

as desired. □

Now, since $|E_{\mathrm{eval}}| \sim \mathrm{Binomial}(\binom{n}{2}, n^{-1/2}/(1 + n^{-1/2}))$, we have

References

[MS16] Montanari, A. and Sen, S. (2015) *Semidefinite Programs on Sparse Random Graphs and their Application to Community Detection*, Proceedings of the forty-eighth annual ACM symposium on Theory of Computing, 2016, pp. 814–827.

[1] Author (Year) *Title*, Publisher